

Satellite Formation Flying Control by Using Only Angle Measurements

Jianqing Li, Liangming Chen, Ming Cao, *Fellow, IEEE*, and Chaoyong Li

Abstract

This paper is to investigate a practical formation control problem of a multi-satellite system, in which a group of small satellites are required to reach and maintain a desired geometric configuration under communication and measurement constraints. We consider each satellite uses its local coordinate systems and senses its neighboring satellites to obtain angle measurements. A satellite formation control scheme using only the angle measurements is proposed for forming the desired configurations. Moreover, the triangular and polygonal configuration cases are discussed, in which the sum of interior angles of polygons is utilized to guarantee the stability of the determined final shape. *Particularly, by using formation rigidity ideas (the formation constructs a multi-point framework which be treated as a rigid body), we propose a formation determination approach to construct a rigid square configuration. It is shown that the absolute position of each satellite and the knowledge of the communication topology among satellites are not needed for the satellite formation control. Moreover, the proposed angle-based formation control law has more robustness against the misalignment of measurement coordinates, since the angle measurements are scalars and sensed with respect to the local coordinate system.* Simulation results are presented to illustrate the performance of the proposed control laws.

Index Terms

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I. INTRODUCTION

Satellite formation flying has been of growing interest to environmental measurement and Earth observation due to its large geographic coverage and rich sensing functionalities. In particular, the technological development of small satellites enables satellite formations to excel in their modularity, low cost, and distributed functionality [1]. In order to take advantage of satellite formations to implement space missions, several formation configurations have been extensively studied for engineering applications, such as a triangular formation for detecting gravitational waves [2] and tracking of a missile launch [3], and a hexagonal regular formation for building large antennas [4].

In satellite formation control, the position-based control approach is widely used to maintain the formation configurations. Satellites measure their own positions and velocities through the global positioning system (GPS) or inertial measurement unit (IMU). The desired formation can be achieved by the feedback control of the individual satellite's absolute position without communication between each other [5]. [With the help of the interact topology technology \(which could be a communication topology or sensing topology\), the displacement-based control, which requires satellites obtain relative positions with respect to their neighbors, is applied to improve the robustness and stability of formation keeping.](#) During the last decade, the displacement-based formation control has been widely applied to satellite formation control that can be either centralized or decentralized [6][7]. To achieve the desired formation configuration, the consensus theory is usually taken to design the formation control law [8][9][10].

In recent years, the bearing-based formation control, which determines the desired formation by sensing the relative bearings with respect to neighbors [11][12], has been extensively studied for multi-agent systems. For satellite formation control, the line of sight (LOS) vector acquired by vision-based sensors and star trackers can be used as the bearing measurements for control. [Gurfil \[13\] studied the cyclic pursuit formation problem that each satellite tracks the LOS to a single neighboring satellite so as to form a cycle formation. Due to each satellite only acquires the information from one neighboring satellite in the cyclic topology, the cyclic satellite formation needs minimum communication connections and measurements to effectively reduce information interaction.](#)

However, the position-based, displacement-based, and bearing-based formation control require the measurements to be conducted in a global coordinate system [14], which means the orientations of the coordinate systems of all satellites are aligned to each other. To relax this restriction, some formation control schemes based on distance or angle measurements are investigated [15][16]. In the distance-based formation control, the agents measure relative positions of their neighbors with respect to their own local coordinate systems. To ensure the uniqueness of the desired formation, the distance rigidity, which can be interpreted as all desired distances are met is sufficient for the maintenance of the distances of any pair of agents, is used to describe the desired configuration without bearings[17]. In particular, the angle-based formation control attracted much attention due to the fact that the angle measurements are cheaper and more accessible [14]. More importantly, angle measurements can be taken as bearings defined and measured in local coordinate systems without a common sense of orientation. Basiri [18] first presented the result on provably stable angle-only formation control for a triangular formation case. Further, Bishop [19] developed a distributed control law for triangular formation control using a mixture of distance and angle information. This mixture makes the agents have heterogeneous sensing capabilities. Since the angle measurements are scalars and are invariant to motions including translations, rotations and scalings, a new angle rigidity scheme was developed to solve the formation control problem by using angle constraints. [20]. Note that the existing angle-based control results have focused on the angle-based control for single-integrator agent models. Thus, it is necessary to investigate more realistic applications of angle-based control.

From the practical point of view, unique features of the small satellite formation are threefold. First, less and lower-cost computational instruments are essential for the satellite formation control due to small satellites are constrained by size and power consumption. Second, the previous studies on formation control requires satellites sense measurements with respect to the global coordinate or have the common orientations of local coordinate. This demands for the precision attitude control of all satellites. Third, there is no enough infrastructure for all-to-all communication among small satellites. Moreover, technical challenges arise from communication delays and topology switching. These arguments motivate us to look for a cheaper and more reliable formation control scheme.

In this paper, we consider the circular satellite formation control problem, where only angle measurements are available to each satellite. The angle measurements can be obtained by star trackers or cameras[21][22]. Moreover, a satellite formation control scheme considered the

practical dynamics is proposed, in which the knowledge of the position of neighbouring satellite and communication topology are unnecessary to construct the formation control law. In particular, the orientations of the local measurement coordinate systems of all satellites are unnecessary to be aligned to each other.

The contributions of this paper are summarized as follow: 1) In our proposed formation control scheme, only the angle measurements are required to construct the control law, which implies that an individual satellite does not need to acquire its relative position information with respect to other satellites. 2) Compared with the bearing-based formations where all satellites should have a common coordinate frame or a preset coordinate frame for bearing measurements, the proposed angle-based formation control law is robust to the misalignment in satellites' measurement coordinate frames, i.e., each satellite is allowed to use different coordinate frames for direction measurements. 3) For the angle-based formation control, we provide a specific construction for the square configuration, in which the square formation can be uniquely determined by only sensing one angle measurement of neighbors.

The remainder of this paper is organized as follows. Section 2 introduces the satellite formation control problem. Section 3 presents the main results with different configurations. Section 4 shows the simulation results examining the performance of the proposed formation control.

II. PROBLEM FORMULATION

In this paper, we investigate the relative motion control of a group of n satellites flying on orbits relative to a circular reference orbit such that a satellite formation is obtained and maintained. The dynamic equations of relative motions are expressed in the Local Vertical Local Horizontal (LVLH) frame $o-xyz$ [23], as shown in Fig.1. The reference origin of this coordinate system is set on the circular reference orbit.

Let $I_n \in \mathbb{R}^{n \times n}$, $0_{m \times n} \in \mathbb{R}^{m \times n}$, and $1_n \in \mathbb{R}^{n \times 1}$ be the identity matrix, zero matrix, and column vector with all ones respectively. Let $\|\cdot\|$ denote the Euclidean norm of a vector.

A. Circular formation control

For the n satellites in 3D, let $p_i \in \mathbb{R}^3$ and $v_i \in \mathbb{R}^3$ be the position and velocity of satellite $i \in \{1, 2, \dots, n\}$ expressed in LVLH. As described in [23], the relative dynamics of the i th

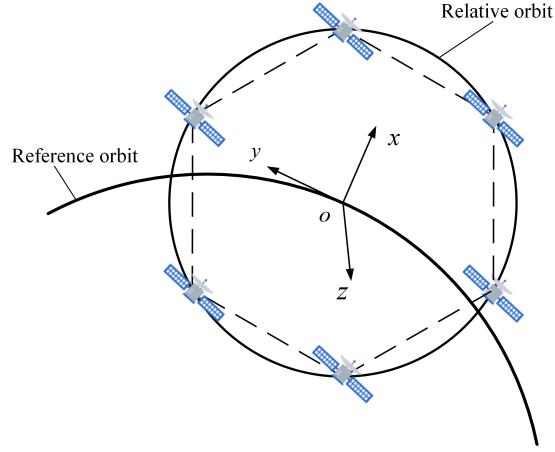


Fig. 1. LVLH coordinate system

satellite are described by

$$\begin{bmatrix} \dot{p}_i \\ \dot{v}_i \end{bmatrix} = \begin{bmatrix} 0_3 & I_3 \\ \Omega_1 & \Omega_2 \end{bmatrix} \begin{bmatrix} p_i \\ v_i \end{bmatrix} + \begin{bmatrix} 0_3 \\ I_3 \end{bmatrix} u_i \quad (1)$$

where

$$\Omega_1 = \begin{bmatrix} 3\omega_r^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\omega_r^2 \end{bmatrix}, \quad \Omega_2 = \begin{bmatrix} 0 & 2\omega_r & 0 \\ -2\omega_r & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\omega_r \in \mathbb{R}$ denotes the constant angular rate of the circular reference orbit, and $u_i \in \mathbb{R}^3$ denotes the control input.

Using the above relative motion model, we will show that when the satellites' initial conditions satisfy some technical relations, each satellite's motion forms a stable relative circular orbit with respect to the reference origin[23]. The following assumptions are made for all satellites.

Assumption 1. *In a typical flying setting, all satellites are initially on their relative circular flying orbits with different radii and maintain the same direction of angular momentum of relative orbits.*

Assumption 2. *The satellites do not communicate with each other and are each equipped with a sensor which provides relative line of sight (LOS) measurements only.*

Assumption 3. The initial satellite positions are separate and noncollinear, i.e. $\cos\langle p_{ij}(0), p_{ik}(0) \rangle \neq \pm 1, \forall i \neq j \neq k$, where $p_{ik} = p_k - p_i$, and $\langle \cdot \rangle$ denotes the subtended angle formed by the corresponding two vectors.

Assumption 1 ensures that all satellites have the same direction of motion (clockwise or counterclockwise direction) and their relative orbits are coplanar. Assumption 2 implies that there is no requirement on the communication topology. For the local sensing, satellites sense the relative LOS angles of their neighboring satellites with respect to their local coordinate systems. The noncollinear condition of Assumption 3 guarantees the formation will not become collinear for $t > 0$, and the collision will not happen for $t > 0$. The analysis of the collinearity and collision can be found in [24].

Under Assumptions 1-3, satellite i measures the LOS angle $\theta_{ij} \in [0, \pi]$ with respect to its neighboring satellite j in its own coordinate frame as shown in Fig.2. The origin of the measurement coordinate system is assumed to be fixed at the center of mass of the satellite. The x_{s_i} , y_{s_i} , and z_{s_i} axes form a right-hand frame, where z_{s_i} axis is parallel to the angular momentum of relative orbits. Then, the angle α_i in the x_{s_i} - y_{s_i} plane can be obtained by as follows:

$$\alpha_i = \theta_{ij} - \theta_{ik} \quad (2)$$

where θ_{ij} and θ_{ik} are measured counterclockwise from the x_{s_i} axis of the local coordinate frame.

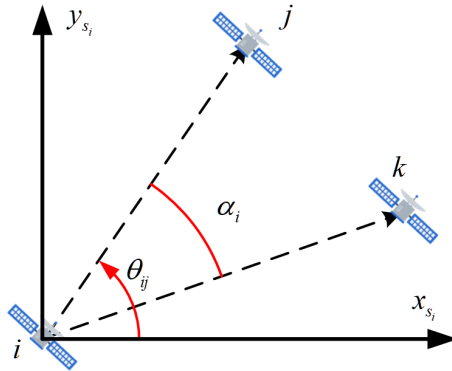


Fig. 2. LOS angle measurements

In this work, we focus on a desired geometric configuration in which all satellites operate around the reference origin with the identical radius, maintain a constant distance from each other, and thus form the circular formation of satellites. The circular formation problem is formally defined as follows.

Problem 1 (Angle-based Circular Formation Control). *Consider a group of $n \geq 3$ satellites with relative dynamics defined in (1) under Assumptions 1-3. Find a formation control law for each satellite using only relative LOS angle measurements such that the circular formation is achieved.*

This problem has not been solved before due to the requirement that the control law only employs angle measurements taken by vision-based sensors in local coordinates and also due to the specific dynamics of satellites. Each satellite knows its own position and velocity, and does not consider the constraints of any other satellite when executing its own control law.

B. Angle-based formation configuration

As shown in Fig.1, we employ a convex polygon to construct the desired circular formation configuration for this group of $n \geq 3$ satellites. In the convex polygon formation, satellites spread in a common circle and measure only the relative LOS vector towards their neighbouring satellites. Then, the interior angle α_i defined in Fig.2 can be calculated by

$$\cos \alpha_i = z_{ij}^T z_{ik} \quad (3)$$

where $z_{ij} = (p_j - p_i) / \|p_j - p_i\| \in \mathbb{R}^3$ is the LOS bearing between satellites i and j .

Let $\alpha_i^* \in (0, \pi)$ be the desired steady-state angles of the convex polygon. Since α_i^* are the interior angles of the desired convex polygon, it follows that $\sum_{i=1}^n \alpha_i^* = (n - 2)\pi$. Then, the objective of the angle-based formation control problem can be converted to the realization of the desired angle constraints as follows:

$$\lim_{t \rightarrow +\infty} (\alpha_i(t) - \alpha_i^*) = 0, \quad (4)$$

For example, in our desired configuration defined by the convex polygon construction, a group of three satellites can form a triangular formation in a common circle. Then, the triangular

formation can be completely determined by α_i^* , $i \in \{1, 2, 3\}$. In fact, there is a redundant angle for this configuration determination; namely, once α_1^* and α_2^* are fixed, it holds that $\alpha_3^* = \pi - \alpha_1^* - \alpha_2^*$.

For a group of four satellites, the circular formation becomes the square configuration as shown in Fig.3(a). In this example, the square configuration in Fig.3(a) is the desired formation; however, the rectangular configuration in Fig.3(b) has the same interior angles $\alpha_i^* = \pi/2$, $i \in \{1, 2, 3, 4\}$, as the realized formation. As a result, both of the configurations in Fig.3(a)-(b) are possible formation solutions to (4). This implies that the configurations in Fig.3 are not angle rigid [25] because they both have the same interior angles and radii with respect to the reference origin. The solution to the square formation control will be discussed later.

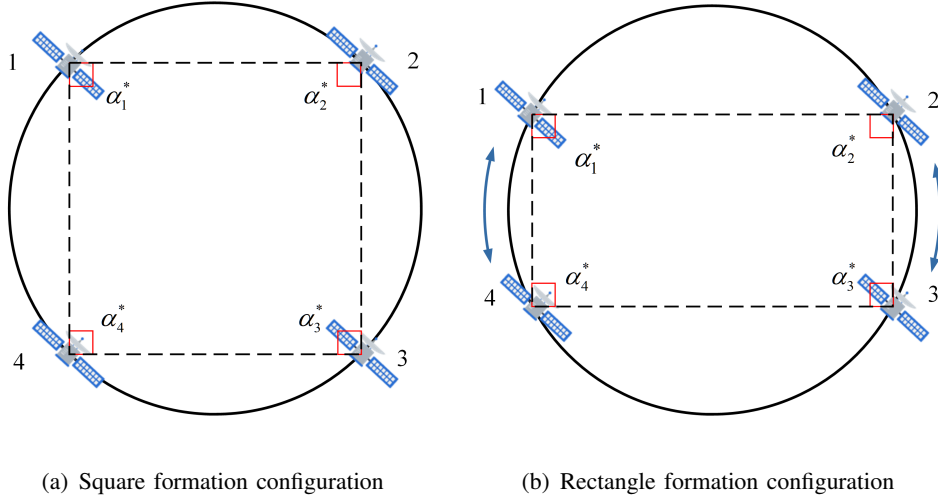


Fig. 3. Two configurations with a group of four satellites

III. MAIN RESULTS

In this section, the solutions to the circular formation control problem with different configurations are presented.

A. Triangular formation control

Consider that a group of $n = 3$ satellites initially operate in their own circular orbits with different radii. To solve Problem 1 by using triangular formation control, the formation control laws for satellite 1 to 3 are proposed to be

$$u_i = \dot{v}_i^* - \Omega_1 p_i - \Omega_2 v_i - k_1(\alpha_i - \alpha_i^*)(z_{i(i+1)} + z_{i(i-1)}) - k_2(v_i - v_i^*) \quad (5)$$

where k_1 and k_2 are positive scalar gains, α_i^* is the desired angle and $v_i^* \in \mathbb{R}^3$ is the desired velocity of satellite i . $z_{i(i+1)} = z_{31}$ when $i = 3$ and $z_{i(i-1)} = z_{13}$ when $i = 1$.

In the relative circular orbit, satellite i 's desired velocity and its derivative can be given by

$$\begin{aligned} v_i^* &= \omega_d \times p_i \\ \dot{v}_i^* &= \omega_d \times \dot{p}_i = \omega_d \times v_i \end{aligned} \quad (6)$$

where $\omega_d = [\pm\sqrt{3}/2, 0, \pm 1/2]^T \omega_r \in \mathbb{R}^3$ is the prearranged angular velocity vector, which can be calculated by $v_d(t) = \omega_d \times p_d(t)$, where $v_d(t)$ and $p_d(t)$ are analytical solutions to the relative circular orbit and can be found in [23]. The sign of ω_d depends on the initial direction of the angular momentum of the satellite motion. It is easy to see that the angular rate of the reference orbit determines the angular velocity of the flying satellites. As a result, once the reference orbit is determined, the flying orbital periods of all satellites are identical with respect to the reference center.

Then, to solve Problem 1, we first check the dynamics described by interior angles. From (3), the angle dynamics of each satellite can be given by

$$\begin{aligned} \dot{\alpha}_1 &= -\frac{[Z_{13}(v_3 - v_1)]^T z_{12}}{l_{13} \sin \alpha_1} - \frac{z_{13}^T Z_{12}(v_2 - v_1)}{l_{12} \sin \alpha_1} \\ \dot{\alpha}_2 &= -\frac{[Z_{23}(v_3 - v_2)]^T z_{21}}{l_{23} \sin \alpha_2} - \frac{z_{23}^T Z_{21}(v_1 - v_2)}{l_{21} \sin \alpha_2} \\ \dot{\alpha}_3 &= -\frac{[Z_{31}(v_1 - v_3)]^T z_{32}}{l_{31} \sin \alpha_3} - \frac{z_{31}^T Z_{32}(v_2 - v_3)}{l_{32} \sin \alpha_3} \end{aligned} \quad (7)$$

where $Z_{ij} = I_3 - z_{ij}z_{ij}^T$, and l_{ij} is the distance between satellites i and j .

Define $e_i = \alpha_i - \alpha_i^*$ and $\delta_i = v_i - v_i^*$, then the angle error dynamics can be rewritten as follows

$$\begin{aligned} \dot{e}_1 &= (z_{12}^T P_{13} + z_{13}^T P_{12})\delta_1 - z_{13}^T P_{12}\delta_2 - z_{12}^T P_{13}\delta_3 \\ \dot{e}_2 &= -z_{23}^T P_{21}\delta_1 + (z_{21}^T P_{23} + z_{23}^T P_{21})\delta_2 - z_{21}^T P_{23}\delta_3 \\ \dot{e}_3 &= -z_{32}^T P_{31}\delta_1 - z_{31}^T P_{32}\delta_2 + (z_{32}^T P_{31} + z_{31}^T P_{32})\delta_3 \end{aligned} \quad (8)$$

where

$$P_{ij} = \frac{Z_{ij}}{l_{ij} \sin \alpha_i} \in \mathbb{R}^{3 \times 3} \quad (9)$$

Now, the convergence of formation control law (5) is analyzed as follows.

Theorem 1. Consider a group of three satellites with the relative dynamics given by (1) under Assumption 1-3. If the initial errors $e_i(0)$ and $\delta_i(0)$, for $i = 1, 2, 3$, locate in a neighborhood

$\mathbb{U}$ of the origin, then the triangular formation control law (5) ensures that $\alpha_i(t) \rightarrow \alpha_i^*$ and $v_i(t) \rightarrow v_i^*(t)$ asymptotically as $t \rightarrow +\infty$.

Proof. With the formation control laws (5), the velocity error dynamics can be expressed as follows

$$\dot{\delta}_i = -k_1(\alpha_i - \alpha_i^*)(z_{i(i+1)} + z_{i(i-1)}) - k_2(v_i - v_i^*) \quad (10)$$

Particularly, since $\alpha_1 + \alpha_2 + \alpha_3 = \pi$, the satellite 3's velocity error dynamics can be rewritten as

$$\dot{\delta}_3 = -k_1(\alpha_1^* - \alpha_1 + \alpha_2^* - \alpha_2)(z_{31} + z_{32}) - k_2(v_3 - v_3^*) \quad (11)$$

Since $\alpha_1 + \alpha_2 + \alpha_3 = \pi$, we use the reduced angle error $e_r = [\alpha_1 - \alpha_1^*, \alpha_2 - \alpha_2^*]^T$ to construct the angle error dynamics. Then within a neighborhood of the origin $\mathbb{U}$, the overall error dynamics can be obtained by linearization of (8), (10), and (11) about the origin

$$\begin{bmatrix} \dot{e}_r \\ \dot{\delta} \end{bmatrix} = H(\alpha^*, v^*) \begin{bmatrix} e_r \\ \delta \end{bmatrix} = \begin{bmatrix} 0_2 & H_1 \\ H_2 & H_3 \end{bmatrix} \begin{bmatrix} e_r \\ \delta \end{bmatrix} \quad (12)$$

where

$$H_1 = \begin{bmatrix} z_{12}^T P_{13} + z_{13}^T P_{12} & -z_{13}^T P_{12} & -z_{12}^T P_{13} \\ -z_{23}^T P_{21} & z_{21}^T P_{23} + z_{23}^T P_{21} & -z_{21}^T P_{23} \end{bmatrix} \bigg|_{\substack{e_r=0 \\ \delta=0}}$$

$$H_2 = \begin{bmatrix} -k_1(z_{12} + z_{13}) & 0 \\ 0 & -k_1(z_{21} + z_{23}) \\ k_1(z_{31} + z_{32}) & k_1(z_{31} + z_{32}) \end{bmatrix} \bigg|_{\substack{e_r=0 \\ \delta=0}}$$

$$H_3 = -k_2 I_9$$

To analyze the convergence of the formation, we construct a rotational LVLH frame that rotates around the constant angular momentum of relative orbits with the angle velocity ω_d . It holds that

$p_i = R(\omega_d t) \bar{p}_i$, where $R(\omega_d t)$ is the orthogonal rotation matrix. Therefore, $\bar{v}_i = \omega_d \times \bar{p}_i = R^T v_i$,

and we have the following transformation:

$$\begin{bmatrix} e_r \\ \bar{\delta} \end{bmatrix} = \begin{bmatrix} I_2 & 0_{2 \times 9} \\ 0_{9 \times 2} & R_k^T \end{bmatrix} \begin{bmatrix} e_r \\ \delta \end{bmatrix} \quad (13)$$

where $R_k = I_3 \otimes R(\omega_d t)$, $\otimes$ is the Kronecker product, and $(\bar{\cdot})$ denotes the vector expressed in the rotational LVLH frame.

Then, Eq.(12) can be transformed to the following linear time-invariant system:

$$\begin{bmatrix} \dot{e}_r \\ \dot{\bar{\delta}} \end{bmatrix} = \bar{H}(\alpha^*, \bar{p}^*) \begin{bmatrix} e_r \\ \bar{\delta} \end{bmatrix} \quad (14)$$

where $\bar{p}^* = [\bar{p}_1^{*T}, \bar{p}_2^{*T}, \dots, \bar{p}_n^{*T}]^T$, $\bar{p}_i^{*T}$ is the stable position vector, and $\bar{H}$ is the constant matrix of H by replacing its p_i^* with the rotation transformation.

Let λ be an eigenvalue of $\bar{H}$. Using a formula for the determinant of a partitioned matrix, we can obtain the characteristic equation associated with $\bar{H}$ as

$$\begin{aligned} & \det(\lambda I - \bar{H}(\alpha^*, v^*)) \\ &= |\lambda I_9 - \bar{H}_3| |\lambda I_2 - \bar{H}_1(\lambda I_9 - \bar{H}_3)^{-1} \bar{H}_2| \\ &= (\lambda + k_2)^9 |\lambda I_2 - \frac{1}{\lambda + k_2} \bar{H}_1 \bar{H}_2| \\ &= (\lambda + k_2)^7 [\lambda^4 + 2k_2 \lambda^3 + (k_2^2 - k_1 q_1 - k_1 q_4) \lambda^2 \\ & \quad - (q_1 k_1 k_2 + q_4 k_1 k_2) \lambda + k_1^2 (q_1 q_4 - q_2 q_3)] \end{aligned} \quad (15)$$

where

$$\begin{aligned} q_1 &= -\frac{\sin \alpha_1^*}{l_{13}^*} - \frac{\sin \alpha_1^*}{l_{12}^*} - \frac{\sin \alpha_3^*}{l_{13}^*} \\ q_2 &= \frac{\sin \alpha_2^*}{l_{12}^*} - \frac{\sin \alpha_3^*}{l_{13}^*} \\ q_3 &= \frac{\sin \alpha_1^*}{l_{21}^*} - \frac{\sin \alpha_3^*}{l_{23}^*} \\ q_4 &= -\frac{\sin \alpha_2^*}{l_{23}^*} - \frac{\sin \alpha_2^*}{l_{21}^*} - \frac{\sin \alpha_3^*}{l_{23}^*} \end{aligned}$$

Now, we pay attention to the coefficients of the second polynomial roots of (15) by using the condition of Routh-Hurwitz criteria. Note that $q_1 < 0$ and $q_4 < 0$, hence all coefficients of the second polynomial are positive and we obtained the following set of inequalities:

$$2k_2(k_2^2 - k_1q_1 - k_1q_4) + k_1k_2q_1 + k_1k_2q_4 = 2k_2^3 - k_1k_2q_1 - k_1k_2q_4 > 0 \quad (16)$$

$$-(q_1k_1k_2 + q_4k_1k_2)[k_2(2k_2^2 - k_1q_1 - k_1q_4)] - 4k_1^2k_2^2(q_1q_4 - q_2q_3) > 0 \quad (17)$$

where inequality (17) is calculated in Appendix A. By examining the above Routh-Hurwitz conditions for the fourth-order polynomial [26], the second polynomial roots of Eq.(15) have negative real parts so that $\bar{H}$ is Hurwitz.

According to Theorem 4.7 of [27] and Eq.(13), it can be concluded that e_r and δ asymptotically converge to zero when the initial errors locate in $\mathbb{U}$. $\square$

Remark 1. It should be pointed out that the proposed formation control law has fewer requirements than the existing results on distributed formation control. For satellite formation systems, the communication delay and topology constraints on formation process are not considered. For each individual satellite, there is no need for the measurement of the position and velocity of neighboring satellites; in other words, the communication among satellites is unnecessary.

Remark 2. Note that the triangular configuration has redundant angle information for the configuration determination. Since $\alpha_1 + \alpha_2 + \alpha_3 = \pi$, we can assume that satellite 1 maintains its own relative orbit, and then satellite 2 and satellite 3 execute the triangular formation control laws to construct the desired configuration with satellite 1. In this way, the relative flying radii of all satellites can be determined to be the flying radius of satellite 1.

Remark 3. We use contradiction to explain that the formation will not become collinear under (5) and Assumption 3. According to the Hurwitz of H , there exist positive constants ρ and λ

such that the following inequality holds when the initial states are in the neighborhood of the desired states.

$$\|\sigma_r(t)\| \leq \rho \|\sigma_r(0)\| e^{-\lambda t}, \quad \forall t \geq 0 \quad (18)$$

where $\sigma_r = [e_r^T, \delta^T]^T$.

Since $e_1^2 \leq \|e_r\|^2 \leq \|\sigma_r(t)\|^2$ and $e_1 + e_2 + e_3 = 0$, we have

$$|e_i| \leq \rho \|\sigma_r(0)\|, \quad i \in \{1, 2\} \quad (19)$$

$$|e_3| \leq 2\|e_r\| \leq 2\rho \|\sigma_r(0)\|$$

Therefore

$$\begin{aligned} \alpha_i^* - \rho \|\sigma_r(0)\| &\leq \alpha_i(t) \leq \alpha_i^* + \rho \|\sigma_r(0)\|, \quad i \in \{1, 2\} \\ \alpha_3^* - 2\rho \|\sigma_r(0)\| &\leq \alpha_3(t) \leq \alpha_3^* + 2\rho \|\sigma_r(0)\| \end{aligned} \quad (20)$$

Because α^* is bounded away from zero and π , there must be a positive time T' such that (20) holds for $t \in [0, T']$. Let T^* be the first time that the collinearity happens in $[T', \infty)$. Then, $\alpha_i(T^*)$ will be zero or π . But there is no collinearity happens in $[0, T^*)$, in view of (20), $\alpha_i(t)$ is bounded away from zero or π which implies a contradiction. Therefore, T^* can be extended to infinity. This proves that no collinearity will happen in $[0, \infty)$ given that the initial state is reasonably close to the desired state.

Remark 4. A situation may arise in which there exists a misalignment angle β in satellite i 's local measurement coordinate. This may result from the installation error of sensors. As shown in Figure 4, the black arrows (solid) denote the original measurement coordinate Σ_s which is aligned with the other satellites' coordinate frames, and the black arrows (dashed) denote the misalignment measurement coordinate Σ'_s .

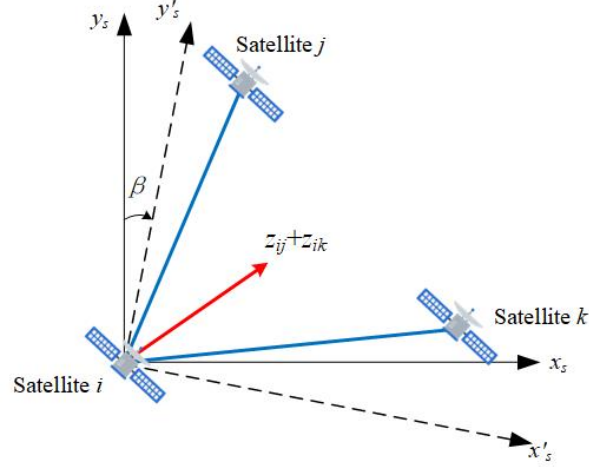


Fig. 4. Local measurement coordinate with misalignment

Then, the control component $z_{ij} + z_{ik}$ described in Σ'_s becomes $R(\beta)(z_{ij} + z_{ik})$. Using this new control component, the formation control law can be described as

$$u_i = \dot{v}_i^* - \Omega_1 p_i - \Omega_2 v_i - k_1(\alpha_i - \alpha_i^*)R(\beta)(z_{i(i+1)} + z_{i(i-1)}) - k_2(v_i - v_i^*) \quad (21)$$

Note that although the bearing vector is affected by the misalignment, the angle measurement is not affected by the misalignment, i.e. α_i is measured precisely. In fact, the sum of the two bearings $z_{ij} + z_{ik}$ points towards the bisector of the interior angle α_i of satellite i , if there is no misalignment. When there is a small misalignment, the pointing direction is still along the interior angle α_i . Thus, our proposed angle-based formation control law is said to be robust since each satellite is only required to move toward the interior of the formation at some direction.

B. Polygonal formation control

In this subsection, we extend the formation control law to solve the polygonal formation control problem. The desired formation configuration is shown in Fig.5 and the corresponding assumption is given as follows.

Assumption 4. *The desired polygonal formation configuration is convex and all satellites have the same flying radius with respect to reference origin.*

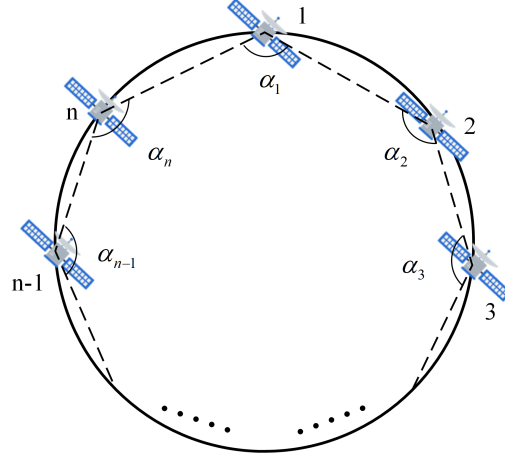


Fig. 5. Polygonal formation configuration

In the convex polygonal formation configuration ($n > 3$), the sum of all interior angles satisfies $\sum_{i=1}^n \alpha_i = (n - 2)\pi$. To construct the desired formation, we propose the following polygonal formation control law:

$$u_i = \dot{v}_i^* - \Omega_1 p_i - \Omega_2 v_i - \frac{k_1(\alpha_i - \alpha_i^*)}{\sin \alpha_i^*} (z_{i(i+1)} + z_{i(i-1)}) - k_2(v_i - v_i^*) \quad (22)$$

The following presents the convergence analysis under the proposed formation control law (22).

Theorem 2. *Consider a group of n satellites with relative dynamics given by (1) under Assumption 1-4. If the initial errors $e_i(0)$ and $\delta_i(0)$, for $i = 1, 2, \dots, n$, locate in a neighborhood $\mathbb{U}$ of the origin, then the polygonal formation control law (22) ensures that $\alpha_i(t) \rightarrow \alpha_i^*$ and $v_i(t) \rightarrow v_i^*(t)$ asymptotically as $t \rightarrow +\infty$.*

Proof. First, with the control law (22), the closed-loop angle dynamics of each satellite can be

given by

$$\dot{e}_i = (z_{i(i+1)}^T P_{i(i-1)} + z_{i(i-1)}^T P_{i(i+1)})\delta_i - z_{i(i+1)}^T P_{i(i-1)}\delta_{i-1} - z_{i(i-1)}^T P_{i(i+1)}\delta_{i+1} \quad (23)$$

where $P_{ij} \in \mathbb{R}^{3 \times 3}$ is defined in (9).

Denote $e = [e_1, e_2, \dots, e_n]^T \in \mathbb{R}^n$ and $\delta = [\delta_1^T, \delta_2^T, \dots, \delta_n^T]^T \in \mathbb{R}^{3n}$, then the dynamics of error system are given by

$$\begin{bmatrix} \dot{e} \\ \dot{\delta} \end{bmatrix} = A(\alpha^*) \begin{bmatrix} e \\ \delta \end{bmatrix} = \begin{bmatrix} 0 & B \\ C & -K_2 \end{bmatrix} \begin{bmatrix} e \\ \delta \end{bmatrix} \quad (24)$$

where

$$B = \begin{bmatrix} \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & 0 & B_{i(i-1)} & B_{ii} & B_{i(i+1)} & 0 & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \end{bmatrix}$$

$$B_{i(i-1)} = -z_{i(i-1)}^T P_{i(i+1)} \in \mathbb{R}^{1 \times n}$$

$$B_{i(i+1)} = -z_{i(i+1)}^T P_{i(i-1)} \in \mathbb{R}^{1 \times n}$$

$$B_{ii} = -B_{i(i-1)} - B_{i(i+1)} \in \mathbb{R}^{1 \times n}$$

$$C = -k_1 \text{diag} \left\{ \frac{z_{i(i-1)} + z_{i(i+1)}}{\sin \alpha_i^*} \right\}, i = 1, 2, \dots, n$$

Similarly, we can obtain the characteristic equation associated with A as

$$\begin{aligned} & \det(\lambda I - A) \\ &= |\lambda I + K_2| |\lambda I - B(\lambda I + K_2)^{-1} C| \\ &= (\lambda + k_2)^{3n} \left| \lambda I - \frac{1}{\lambda + k_2} BC \right| \\ &= (\lambda + k_2)^{2n+1} |\lambda(\lambda I + K_2) - BC| \end{aligned} \quad (25)$$

where

$$BC = \begin{bmatrix} -g_{1n} - g_{12} & g_{12} & 0 & 0 & \cdots & g_{1n} \\ g_{21} & -g_{21} - g_{23} & g_{23} & 0 & \cdots & 0 \\ 0 & g_{32} & -g_{32} - g_{34} & g_{34} & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ g_{n1} & \cdots & \cdots & 0 & g_{n(n-1)} & -g_{n1} - g_{n(n-1)} \end{bmatrix}$$

$$g_{ij} = \frac{k_1}{l_{ij}}$$

Note that the determinant $|\lambda(\lambda I + K_2) - BC|$ is equivalent to the characteristic equation associated with

$$M = \begin{bmatrix} 0 & I \\ BC & -K_2 \end{bmatrix} \quad (26)$$

Because BC has zero row sums and is a symmetric matrix, M has exactly one zero eigenvalue and all other eigenvalues have negative real parts. Moreover, A has exactly one zero eigenvalue and all other eigenvalues have negative real parts.

Define $\sigma(t) = [e^T, \delta^T]^T$, the solution of (24) to the initial state $\sigma(0)$ is given by

$$\sigma(t) = \sum_{k=1}^n e^{\lambda_k t} \mu_k s_k^T \sigma(0) \quad (27)$$

where $\mu_k \in \mathbb{R}^{4n}$ and $s_k^T \in \mathbb{R}^{1 \times 4n}$ are right eigenvector and left eigenvector of A , respectively.

Assume that $\lambda_1 = 0$ It then follows from (27) that

$$\lim_{t \rightarrow +\infty} \sigma(t) = \mu_1 s_1^T \sigma(0) \quad (28)$$

where μ_1 and s_1^T are the right eigenvector and left eigenvector associated with zero eigenvalue, respectively.

Let $\mu_1 = [\mu_p^T, \mu_q^T]^T$ and $s_1^T = [s_p^T, s_q^T]$, where μ_p and s_p are $n \times 1$ column vectors, it follows that

$$\begin{bmatrix} 0 & B \\ C & -K_2 \end{bmatrix} \begin{bmatrix} \mu_p \\ \mu_q \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}_{4n \times 1} \quad (29)$$

$$[s_p^T, s_q^T] \begin{bmatrix} 0 & B \\ C & -K_2 \end{bmatrix} = [0, 0]_{1 \times 4n} \quad (30)$$

From (30), it follows that $s_p^T B - s_q^T K_2 = 0$, one can obtain by expanding

$$\sum_{i=1}^n s_p^i b^{ij} + k_2 s_q^j = 0, \quad j = 1, 2, \dots, 3n \quad (31)$$

where b^{ij} is the entry on the i th row and j th column of B , s_p^i and s_q^i are the i th entry of s_p and s_q , respectively.

Note that all the column sums of B are zero. Therefore, by solving the above algebraic equations, we have

$$s_p^1 = s_p^2 = \dots = s_p^n = \gamma \quad (32)$$

$$s_q^1 = s_q^2 = \dots = s_q^{3n} = 0$$

where γ is a constant. Then, we have

$$\mu_1 s_1 = \begin{bmatrix} \mu_p s_p^T & 0_{n \times 3n} \\ \mu_q s_q^T & 0_{3n \times 3n} \end{bmatrix} = \gamma \begin{bmatrix} \mu_p 1_n^T & 0_{n \times 3n} \\ \mu_q 1_n^T & 0_{3n \times 3n} \end{bmatrix} \quad (33)$$

From the above results, it is obvious that

$$\begin{aligned} \lim_{t \rightarrow +\infty} \sigma(t) &= \gamma \begin{bmatrix} \mu_p 1_n^T & 0_{n \times 3n} \\ \mu_q 1_n^T & 0_{3n \times 3n} \end{bmatrix} \begin{bmatrix} e(0) \\ \delta(0) \end{bmatrix} \\ &= \gamma \mu_1 \sum_{i=1}^n e_i(0) \end{aligned} \quad (34)$$

Because $\sum_{i=1}^n e_i(0) = 0$ from Assumption 4, it follows that $\sigma(t)$ converges asymptotically to zero, which completes the proof. $\square$

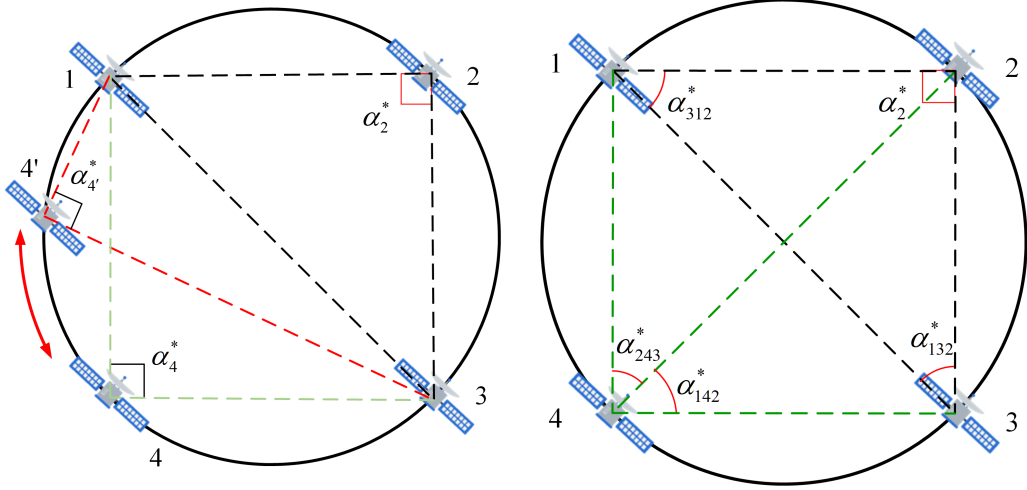
Remark 5. In fact, although the desired polygon formation is achieved, the flying radius of the formation is ambiguous when all satellites converge to the formation stability. To determine the flying radius, by using the rule of the sum of interior angles, the motion scheme for the polygon formation is that one satellite maintains the desired circular orbit without the control input, and its flying radius ($r_1 = \|p_1\|$) is the desired radius r^* . Then, the rest of satellites construct the desired polygon configuration under the polygonal formation control law. As a result, the flying radii of all satellites converge to the desired radius r^* at the steady state.

C. Square formation control

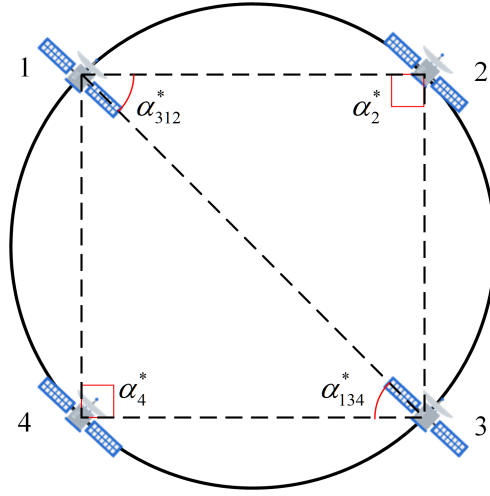
As discussed in the section 2.B and Fig.3, the convex polygon construction is unable to solve circular formation with square configuration. In this subsection, we discuss the construction and control for square formation with four satellites.

Firstly, Fig.6(a) shows a construction scheme that satellites 1-3 construct a triangle configuration $\triangle 123$ (black dashed) with $\alpha_{213}^* = \alpha_{231}^* = \pi/4$ and $\alpha_2^* = \pi/2$, and then satellite 4 senses the LOS angle of satellite 1 and 3 to establish the interior angle $\alpha_4^* = \pi/2$ (green dashed). However, in this case, it is obvious that the satellite 4's position cannot be uniquely determined since α_4^* is also reached (red dashed) when satellite 4 move along the arc $\widehat{143}$. As a result, this construction can not uniquely determine the square configuration. In order to guarantee the angle rigidity under the square formation, we still maintain the triangle configuration $\triangle 123$. Then, satellite 4 can attain two desired angles $\alpha_{142}^* = \alpha_{243}^* = \pi/4$ established with the other three satellites, as shown in Fig.6(b). This construction ensures that the formation is angle rigid with each satellite sensing two angles. However, the increasing number of measured angles will raise the measuring burden of the sensor in this construction case.

Thus, according to the redundant angle of the triangle configuration, we proposed a construction satisfying angle rigidity, in which each satellite only measures one angle with two other satellites. As shown in Fig.6(c), satellites 1 and 3 establish α_{312} and α_{134} to construct the isosceles right triangle $\triangle 123$, and satellite 2 and 4 establish α_2 and α_4 to construct the isosceles



(a) Square formation with sequential construction (b) Square formation with two angles measurements



(c) Square formation with angle rigid construction

Fig. 6. Square formation construction

right triangle $\triangle 341$. Then, the aim of square formation control is to satisfy the following forms:

$$\begin{aligned}
 \lim_{t \rightarrow +\infty} e_{312} &= \lim_{t \rightarrow +\infty} (\alpha_{312} - \pi/4) = 0 \\
 \lim_{t \rightarrow +\infty} e_2 &= \lim_{t \rightarrow +\infty} (\alpha_2 - \pi/2) = 0 \\
 \lim_{t \rightarrow +\infty} e_{134} &= \lim_{t \rightarrow +\infty} (\alpha_{134} - \pi/4) = 0 \\
 \lim_{t \rightarrow +\infty} e_4 &= \lim_{t \rightarrow +\infty} (\alpha_4 - \pi/2) = 0
 \end{aligned} \tag{35}$$

In view of the above objective, the square formation control laws are proposed as follow:

$$\begin{aligned}
u_1 &= T_1 - k_1(\alpha_{312} - \alpha_{312}^*)(z_{12} + z_{13}) \\
u_2 &= T_2 - k_1(\alpha_2 - \alpha_2^*)(z_{21} + z_{23}) \\
u_3 &= T_3 - k_1(\alpha_{134} - \alpha_{134}^*)(z_{31} + z_{34}) \\
u_4 &= T_4 - k_1(\alpha_4 - \alpha_4^*)(z_{43} + z_{41})
\end{aligned} \tag{36}$$

where $T_i = \dot{v}_i^* - \Omega_1 p_i - \Omega_2 v_i - k_2(v_i - v_i^*)$.

The following theorem presents the convergence analysis of the square formation control.

Theorem 3. *Consider a group of four satellites with relative dynamics given by (1) under Assumption 1-3. If the initial errors $e_i(0)$ and $\delta_i(0)$, for $i = 1, 2, 3, 4$, locate in a neighborhood $\mathbb{U}$ of the origin, then the square formation control laws (36) ensure that formation errors $e(t)$ and $\delta(t)$ asymptotically converge to zero.*

Proof. According to the proposed construction in Fig.4(c), the angle error dynamics are given by

$$\begin{aligned}
\dot{e}_{312} &= (z_{12}^T P_{13} + z_{13}^T P_{12})\delta_1 - z_{13}^T P_{12}\delta_2 - z_{12}^T P_{13}\delta_3 \\
\dot{e}_2 &= -z_{23}^T P_{21}\delta_1 + (z_{21}^T P_{23} + z_{23}^T P_{21})\delta_2 - z_{21}^T P_{23}\delta_3 \\
\dot{e}_{134} &= -z_{34}^T P_{31}\delta_1 - z_{31}^T P_{34}\delta_4 + (z_{34}^T P_{31} + z_{31}^T P_{34})\delta_3 \\
\dot{e}_4 &= -z_{43}^T P_{41}\delta_1 + (z_{41}^T P_{43} + z_{43}^T P_{41})\delta_4 - z_{41}^T P_{43}\delta_3
\end{aligned} \tag{37}$$

The velocity error dynamics

$$\begin{aligned}
\dot{\delta}_1 &= -k_1(\alpha_{312} - \alpha_{312}^*)(z_{12} + z_{13}) - k_2(v_1 - v_1^*) \\
\dot{\delta}_2 &= -k_1(\alpha_2 - \alpha_2^*)(z_{21} + z_{23}) - k_2(v_2 - v_2^*) \\
\dot{\delta}_3 &= -k_1(\alpha_{134} - \alpha_{134}^*)(z_{31} + z_{34}) - k_2(v_3 - v_3^*) \\
\dot{\delta}_4 &= -k_1(\alpha_4 - \alpha_4^*)(z_{41} + z_{43}) - k_2(v_4 - v_4^*)
\end{aligned} \tag{38}$$

Let $e = [e_{312}, e_2, e_{134}, e_4]^T$. Then, within a neighborhood of the origin $\mathbb{U}$, the error dynamics

for satellite 1 to 4 can be obtained by linearization about the origin

$$\begin{bmatrix} \dot{e} \\ \dot{\delta} \end{bmatrix} = S(\alpha^*, v^*) \begin{bmatrix} e \\ \delta \end{bmatrix} = \begin{bmatrix} 0 & S_1 \\ S_2 & -K_2 \end{bmatrix} \begin{bmatrix} e \\ \delta \end{bmatrix} \quad (39)$$

where

$$S_1 = \begin{bmatrix} z_{12}^T P_{13} + z_{13}^T P_{12} & -z_{13}^T P_{12} & -z_{12}^T P_{13} & 0 \\ -z_{23}^T P_{21} & z_{21}^T P_{23} + z_{23}^T P_{21} & -z_{21}^T P_{23} & 0 \\ -z_{34}^T P_{31} & 0 & z_{34}^T P_{31} + z_{31}^T P_{34} & -z_{31}^T P_{34} \\ -z_{43}^T P_{41} & 0 & -z_{41}^T P_{43} & z_{41}^T P_{43} + z_{43}^T P_{41} \end{bmatrix} \bigg|_{\substack{e_r=0 \\ \delta=0}}$$

$$S_2 = -k_1 \begin{bmatrix} z_{12} + z_{13} & 0 & 0 & 0 \\ 0 & z_{21} + z_{23} & 0 & 0 \\ 0 & 0 & z_{31} + z_{34} & 0 \\ 0 & 0 & 0 & z_{41} + z_{43} \end{bmatrix} \bigg|_{\substack{e_r=0 \\ \delta=0}}$$

$$K_2 = k_2 I_{12}$$

Similarly, $\bar{S}$ can be acquired by applying the coordinate transformation in the proof of Theorem

1 to Eq.(39). Then, one can obtain the characteristic equation associated with $\bar{S}$ as

$$\det(\lambda I - \bar{S}) = (\lambda + k_2)^9 |\lambda(\lambda I + K_2) - \bar{S}_1 \bar{S}_2| \quad (40)$$

Since $\alpha_{312}^* = \alpha_{134}^* = \pi/4$ and $\alpha_2^* = \alpha_4^* = \pi/2$, it follows that $\bar{S}_1 \bar{S}_2$ can be calculated as

$$\bar{S}_1 \bar{S}_2 = \frac{k_1}{l^*} \begin{bmatrix} -\frac{1+\sqrt{2}}{2} & 1 & \frac{1}{2} & 0 \\ \frac{\sqrt{2}}{2} & -2 & \frac{2+\sqrt{2}}{2} & 0 \\ \frac{1}{2} & 0 & -\frac{1+\sqrt{2}}{2} & 1 \\ \frac{2+\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} & -2 \end{bmatrix} \quad (41)$$

where l^* is the edge length of the desired square configuration.

It can be concluded that the eigenvalues of $\bar{S}_1\bar{S}_2$ have negative real parts by numerical calculation. Then, $\bar{S}$ is Hurwitz stable. Therefore, the formation errors e and δ locally and asymptotically converge to zero. $\square$

IV. SIMULATIONS

In order to illustrate the performance of the proposed formation control laws, several simulations are performed to implement the typical formation configurations mentioned in [3] and [4]. In the relative circular orbit, for each satellite, the required initial conditions for forming a flying circular orbit are given by

$$\begin{aligned} v_{yi}(0) &= -2\omega_r p_{xi}(0), & v_{xi}(0) &= \omega_r p_{yi}(0)/2 \\ p_{zi}(0) &= \sqrt{3}p_{xi}(0), & v_{zi}(0) &= \sqrt{3}v_{xi}(0) \end{aligned} \quad (42)$$

where $\omega_r = 9.96 \times 10^{-4}$ rad/s is determined by the orbit radius of the circular reference orbit, which is selected to be 7378km.

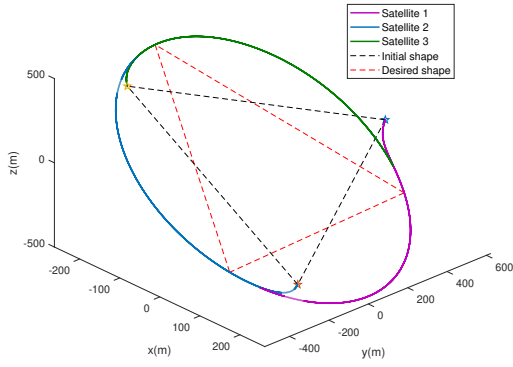
A. Triangular formation case

We first consider the triangular formation case where the three satellites initially are spaced in their own orbits with different radii ($r_1 = 620$ m, $r_2 = 420$ m, $r_3 = 500$ m). Then, let the desired configuration be the equilateral triangle, i.e. $\alpha_1^* = \alpha_2^* = \alpha_3^* = \pi/3$.

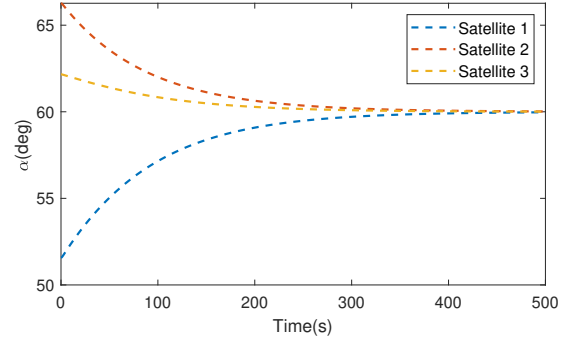
By applying the triangular formation control law (5) with $k_1 = 5$ and $k_2 = 1$, the simulation results are shown in Fig.7. In Fig.7(a), the initial states of the three satellites form an arbitrary triangle (black dashed) and the solid lines depict the trajectories of each satellite as the formation converges to the desired configuration (red dashed). Fig.7(b) shows that the interior angles α_i all tend to $\pi/3$ at the steady state, which demonstrates that the three satellites reach the desired formation configuration.

B. Polygonal formation case

In this case, we consider a hexagonal formation configuration with a group of six satellites. Particularly, in order to illustrate the determination of the flying radius, the motion strategy specified in Remark 5 is carried out. To this end, satellite 1 is settled as the uncontrolled satellite,



(a) Interior angle with triangular formation control

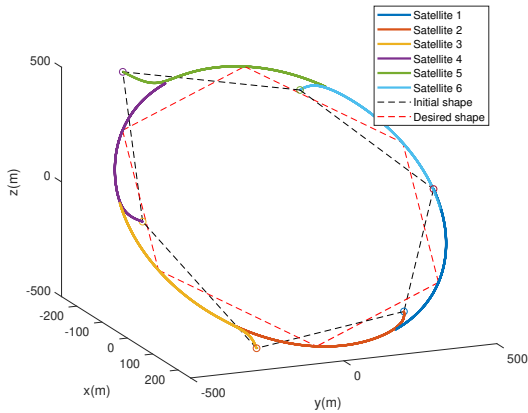


(b) Angle trajectories

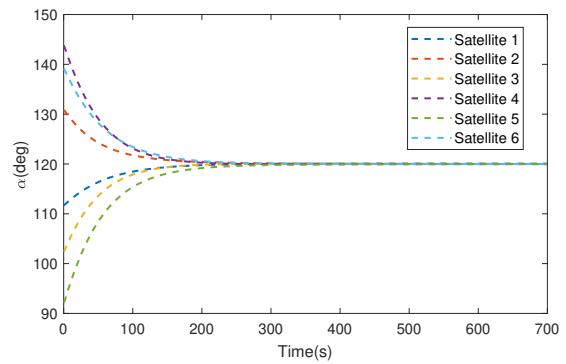
Fig. 7. Three satellites with triangular formation control law

and then satellites 2-6 execute the polygonal formation control law (22) to construct the hexagonal formation configuration.

The simulation results are shown in Fig.8. It can be seen that the motion of satellite 1 remains unchanged (dark blue solid line), while the trajectories of satellite 2-6 are close to the relative circular orbit of satellite 1 and construct a hexagonal configuration (red dashed), i.e. the six satellites are evenly spaced in the final circular formation..



(a) Interior angle with polygonal formation control



(b) Angle trajectories

Fig. 8. Six satellites with polygonal formation control law

C. Square formation case

In this case, we perform the polygonal formation control law (22) ($i = 1, 2, 3, 4$) and the square formation control law (36) to form the desired square formation configuration for comparison. The selection of the control gains are similar to case 1.

The comparison results are shown in Fig.9 and Fig.10. In Fig.9, the satellites converge to a steady formation (red dashed) with desired interior angles ($\alpha_1^* = \alpha_2^* = \alpha_3^* = \alpha_4^* = \pi/2$), but this configuration is a rectangle where $l_{12} = l_{34} = 643\text{m}$ and $l_{14} = l_{23} = 748\text{m}$. Conversely, by executing the square formation control law, the desired interior angles can be achieved ($\alpha_{312} = \alpha_{314} = \pi/4$ and $\alpha_2 = \alpha_4 = \pi/2$) and that the edges of the final configuration are equivalent ($l^* = 1073\text{m}$), which implies that the steady configuration is a square (red dashed). This result demonstrates that our proposed square formation control is applicable.

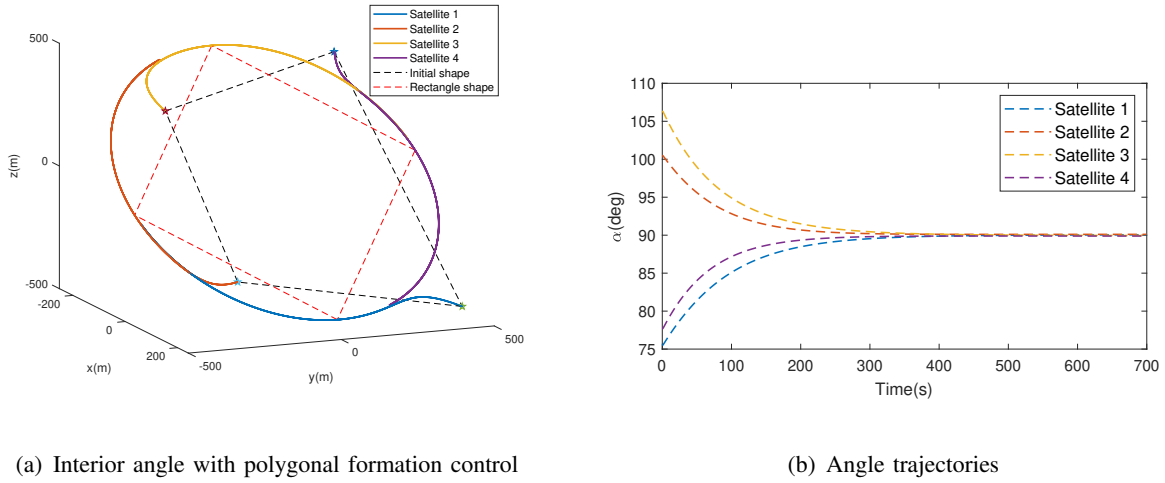
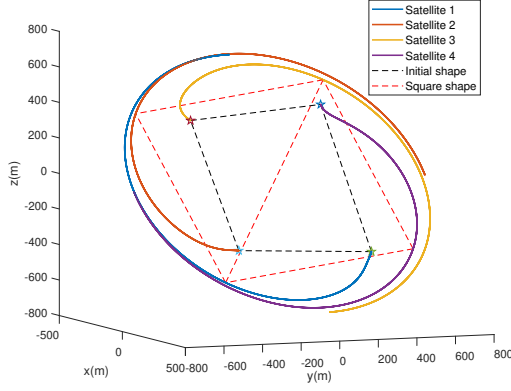


Fig. 9. Four satellites with polygonal formation control law

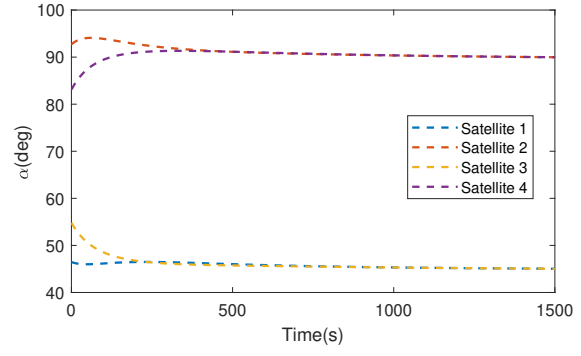
D. Uncertainties case

The purpose of this case is to consider a more realistic simulation when the LOS measurement and the navigation information cannot be precisely obtained.

First, we consider the effect of uncertainties on the relative LOS measurements for the equilateral triangle formation. Suppose that there exists an measuring error 2° in the LOS measurements. The simulation results are shown in Fig.11. To intuitively draw a conclusion, the variations of the radii of the relative circular orbits are depicted in Fig.11(a). It is seen that



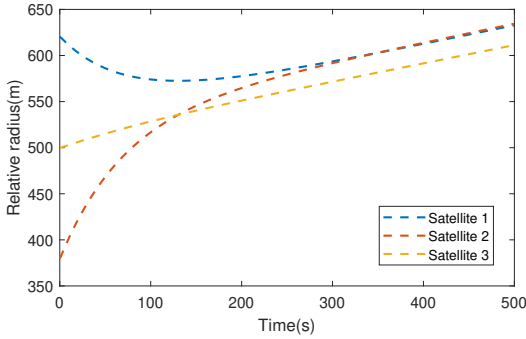
(a) Interior angle with square formation control



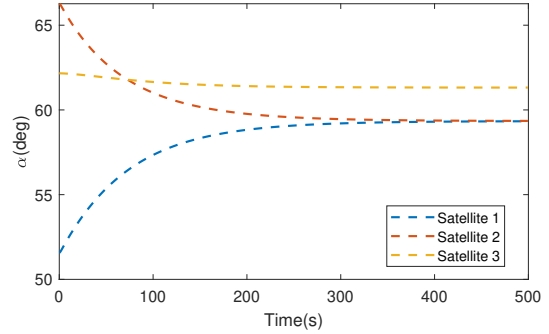
(b) Angle trajectories

Fig. 10. Four satellites with square formation control law

the radii of relative circular orbits are not consensus and increase rapidly, which indicates that all satellites drift away. Comparing to the desired angles of the equilateral triangle formation, it can be seen that the interior angles tend to stability, but the stable angles are not expected. In this sense, we conclude that the desired formation is not achieved.



(a) Radii of the relative circular orbits



(b) Angle trajectories

Fig. 11. Triangular formation with LOS uncertainties

Second, we consider the effect of uncertainties on the absolute navigation sensors. Suppose that the uncertainties of absolute navigation sensors are in order of 10%. The results of the simulation are depicted by Fig.12. From Fig.12(a), we see that the radii of relative orbits roughly reach a consensus. The interior angles virtually converge to the neighborhood around the desired angles, while the angles present in a secular oscillatory, as shown in Fig.12(b). It is clear that

the accuracy of the formation is affected by the uncertainties on the absolute orbital state.

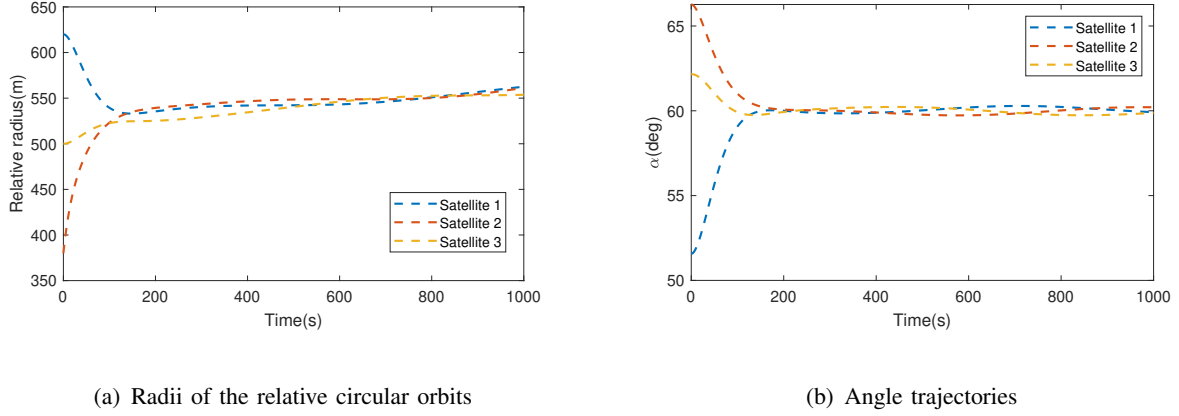


Fig. 12. Triangular formation with navigation uncertainties

V. CONCLUSION

In this paper, we exploit a formation control scheme to deal with the small satellite formation control problem where there is no need for the position measurement of each satellite and communication equipment. For the triangular formation configuration, the desired configuration can be determined by any two satellites attaining expected angles. Furthermore, we extend the result to obtain the polygonal formation configuration. In this case, the determination for the formation flying radius is discussed, in which one satellite's relative orbit is fixed and the rest of the satellites construct the desired formation shape. In addition, in order to address the construction problem for the unique square configuration under the proposed control law, a simple construction is proposed allowing that each satellite only senses one angle. Simulation results demonstrate that all satellites can form the desired formation configurations and maintain their own desired motions.

APPENDIX A

In view of (17), it becomes

$$\begin{aligned}
 & -(q_1 k_1 k_2 + q_4 k_1 k_2)[k_2(2k_2^2 - k_1 q_1 - k_1 q_4)] \\
 & - 4k_1^2 k_2^2 (q_1 q_4 - q_2 q_3) \\
 & = k_2^2 [-2k_1 k_2^2 (q_1 + q_4) + k_1^2 (q_1 - q_4)^2 + 4k_1^2 q_2 q_3]
 \end{aligned} \tag{43}$$

According to the law of sines, we have

$$\frac{l_{23}}{\sin \alpha_1^*} = \frac{l_{13}}{\sin \alpha_2^*} = \frac{l_{12}}{\sin \alpha_3^*} = d \quad (44)$$

so that the last two terms of Eq.(43) can be written as

$$\begin{aligned} & k_1^2(q_1 - q_4)^2 + 4k_1^2q_2q_3 \\ &= k_1^2d^2 \left(\frac{l_{23}}{l_{13}} + \frac{l_{23}}{l_{12}} + \frac{l_{12}}{l_{13}} - \frac{l_{13}}{l_{23}} - \frac{l_{13}}{l_{12}} - \frac{l_{12}}{l_{23}} \right)^2 \\ &+ 4k_1^2d^2 \left(\frac{l_{23}}{l_{12}} - \frac{l_{12}}{l_{23}} \right) \left(\frac{l_{13}}{l_{12}} - \frac{l_{12}}{l_{13}} \right) \end{aligned} \quad (45)$$

Now, there are two cases to determine the positivity of the above equation. First, if l_{12} is the maximal side or minimal side, we have

$$\left(\frac{l_{23}}{l_{12}} - \frac{l_{12}}{l_{23}} \right) \left(\frac{l_{13}}{l_{12}} - \frac{l_{12}}{l_{13}} \right) = \left(\frac{l_{23}^2 - l_{12}^2}{l_{12}l_{23}} \right) \left(\frac{l_{13}^2 - l_{12}^2}{l_{12}l_{13}} \right) > 0 \quad (46)$$

so that $k_1^2(q_1 - q_4)^2 + 4k_1^2q_2q_3 > 0$.

Second, if l_{12} is the middle side, let $a = l_{23}/l_{12} - l_{12}/l_{23}$, $b = l_{13}/l_{12} - l_{12}/l_{13}$, $c = l_{23}/l_{13} - l_{13}/l_{23}$. Then, we have

$$\begin{cases} a \leq 0, b \geq 0, c \leq 0, & l_{13} \geq l_{12} \geq l_{23} \\ a \geq 0, b \leq 0, c \geq 0, & l_{23} \geq l_{12} \geq l_{13} \end{cases} \quad (47)$$

It follows that

$$b(a - c) \geq 0 \quad (48)$$

From (45), we obtain the following inequality

$$\begin{aligned} & k_1^2(q_1 - q_4)^2 + 4k_1^2q_2q_3 \\ &= k_1^2d^2[(a + b - c)^2 + 4ac] \\ &= k_1^2d^2[a^2 + b^2 + c^2 + 2ab + 2ac - 2bc] \\ &= k_1^2d^2[(a + c)^2 + b^2 + 2b(a - c)] \\ &\geq 0 \end{aligned} \quad (49)$$

With the above two cases for l_{12} , one has the inequality (17).

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